CS 610

Past Paper

Final Term prepation

Question: 1

Suppose you are given task to check the connectivity of some website host then which command you will you for this purpose? Also answer the following question in context of your purposed command?

Solution:

1) What will be the default wait time in million second.?

The default wait time for a response from each packet usually 1000 million second.

2) Mention the symbol that will show in case of dropped packets.?

If a packet are dropped, you will trying See a Request time outed. usually message is a period (router) that symbol in some implementation.

3) what will be the maximum value of TTL.?

The maximum value of TTL is 255. This

value can represent the maximum number of hops (router) that the packet can travers before being discarded.

Question: 2

Following table contain diffe IP address. You are required mention the class of each given IP address.

Solution

IP Address	Class of IP addres
105304	A
241.0.01	F
192.168.32.15	C
138.4.4.87	B
237.1.0.9	D

Question: 3

Given below are some feature of IPv4 and IPv6 version. You are required to identify the appropriate version of IP (IPv4/IPv6) against each form. Provide your answer in the given table?

Solution:

Feature	IP version (IPv4/IPv6)
1) Address are 32-bit length.	IP v 4
2) Abuilt IPsec support	IP v 6
3) Ragmentation is done only by sender	IP v 6
4) Broadcast message are avaliable.	IPv4

Question:4

Types of delay when network components take time to forward data over the network medium.?

Answer

The types of delay is a processing delay.

Question:5

IPv4 datagram header field from the given option.?

Date: _____

Answer:

1) Source IP address
2) Destination IP address
3) version 4) Header length
5) Total length

Question: 6

Router matric used by RIP:

Answer:

The Router matric used by RIP is Hop Count.

Question: 7

Two ICMP information message from the given list.?

Answer:

The list is not provided, but two common ICMP information message are:

1) Echo Request
2) Echo Reply

Question: 8

Date:

Subnet mask for the IP address 110.0.1.2/13.

Answer:

The subnet IP address 110.0.1.2/13 is 255.248.00 = 11111111 11111000, 00000000, 00000000
255 248 0 0

Question: 9 **long Question**

Five types of multicast Router protocoal with complete name.?

Answer:

1) Protocol independent Multicast (PIM)
2) Distance vector multicast Routing Protocol (DVMRP)
3) Multicast open shortest path first (MOSPF)
4) Core Based Tree (CBT)
5) Boder Gateway multicast protocol (BGP)

Question: 10

Types of quality service specification of ATM.?

Answer:

2) Sustainable Cell Rate (SCR)

3) Maximum Burst Size (MBS)

Question: 11

Given IP address 210.1.1.2/

Answer:

The number of bits used for Subnetting is 2 (30-28=2) where 28 is number of bits for class C IP address.

Class B:

The class of the IP address is class C.

Subnet mask:

The subnet mask is 255.255.252

11111111, 11111111, 11111111, 11111100 = 255.255.252

128 64 32 16 8 4 2 1

Question: 12

If bridge an intelligence device.? If yes then how.

Answer:

Yes a bridge is considered

Date:

an intelligence device in networking.

1) Learn MAC address
2) Filters traffic 3) Foword intelligency
4) Reduces collisions.

Question: 13

Network adminstrator wants to get periodic updates of routing table.? which routing algorithm would be used there.?

Answer:

The distance routing table would be used.

Reason:

1) Distance vector routing shares routing information with neighboring routers at regualar time intervals.

2) It update the entire routing table periodically even if there are no change.

Question: 14

What are the service

provided by TCP.?

Answer:

1) Connection oriented communication
2) Reliable data transfer
3) Flow control 4) Error detection and Recovery
5) Congestion control 6) Ordered data delivery
7) Full duplex Communication

Question: 15

What is the concept of zero-compression regarding IPv6.?

Answer:

Zero compression is a method of in IPv6 used to shorten long sequence of consecutive zero in an IP address to make it easier to read and write.

Question: 16

What are the function IP multicast abstraction allow an application running on an abitrary computer to do.?

Date:

Answer:

IP multicast Abstraction Allow an application it to perform the following function.

1) Join a Multicast Group
2) Leave a Multicast Group
3) Send data Multicast Group
4) Recieve data Multicast Group

Question: 17

Why server is needed.? Define your answer with the help of example.

Answer:

A server is a powerful computer or system that provide services, data, resources, or program to other computer, known as client over a network. Server for essential for hosting websites, manging data and sending email and more.

Example:

When you open google.com your phone ask the server and the server sends the web page.

Date:

Question: 18

In case of UdP describe endpoint identification with protocool port number.?

Answer:

i) IP Address ii) Port Number

IP Address:

Identifies the device on the network.

Port Number:

Identifies the specific application on that device.

Question: 19

Network administrator system receive some fregments of IP datagram. How does IP Software ressemble fregment that arrive out of oder.?

Answer:

When IP datagram fregment arrive out of oder, the IP software ressembles them using three key field in

the IP header.

1) identification field 2) Fragment offset
3) More Fragment.

Question: 20

If we send a frame frome one machine to another machine, what kind of data is being inside the frame and how can arrive know it.?

Answer:

i) Header ii) playload iii) Trailer

Header:

Include a Source, destination MAC addressess, and field type.

playload:

The actual data (like IP packet ARP message etc.)

Trailer:

Error checking info like CRC.

Question: 21

Which has responsibility

Date:

for the propagation of multicast routing?

Answer:

The responsibility for propagation of multicast routing lies with Multicast routing protocols.

Example:

Multicast routing protocols send one copy of data to many user by building efficient paths.

Question: 22

What are common protocol of TCP Networking?

Answer:

1) Wireshark 2) Nmap 3) ping 4) traceout 5) netstate 6) Telnet 7) SSH 8) FTP

Question: 23

What is table lookup, Message Exchange and closed from computation?

Answer:

Date:

Question: 23

Following the table contain the name of different technologies which are used in computer network. You are required to fill the table with respect to (LAN/WAN) indicating where these technologies are used?

Answer:

Technology	LAN/WAN
Ethernet	LAN
Token Ring	LAN
FDDI	LAN
Framme Relay	WAN
Local Talk	LAN

Question: 24

Consider the IP address 178.200.127.5 and its Correspounding Subnet mask 255.255.255.0 find out the following.

a) The number of bits used for subnetting.

b) Total number of useable hosts in the subnet.

c) Write the CIDR notation for the above Scenario

Answer:

Subnetting bits = 255.255.255.0

= 24

Host bit = 32 - 24 = 8 bit

Total host: $2^h = 2^8 = 256$

Useable host = 256 - 2 = 254

8 = bits used for subnetting

254 = useable host

176.200.127.5 / 24 Ans,

Question: 25

Find the last assignable host address in the block if one of the Subnet ID is 192.1.1.0 /26

Answer: Subnet ID = 26

CIDR = 26

Network bits = 26

Host bit = 32 - 26 = 6

Total IP = 2^6 = 64

64 - 2 = 62

192.1.1.62 Ans,

Date: Class

1 , 127 = A

128, 191 = B

192, 224 = C

225, 239 = D

240, 256 = E

Short

Question:26

Identify the three information message.?

Answer:

1) Echo request /reply
2) Address mask request /reply
3) Router discovery

Question:27

Suppose in computer network of ABC university network execution is given a task to check the status of the network station IP stack by planning the address. which prefix would be used to perform this task.?

Answer:

127.0.0.1 (loop IP address)

Question:28

An internet may have network with different MTU. Suppose router detect datagram

larger than network MTU, and then split into pieces. after spliting the datagram into pices. you are request to write the name of the technique that will be used to split datagram into pieces.?

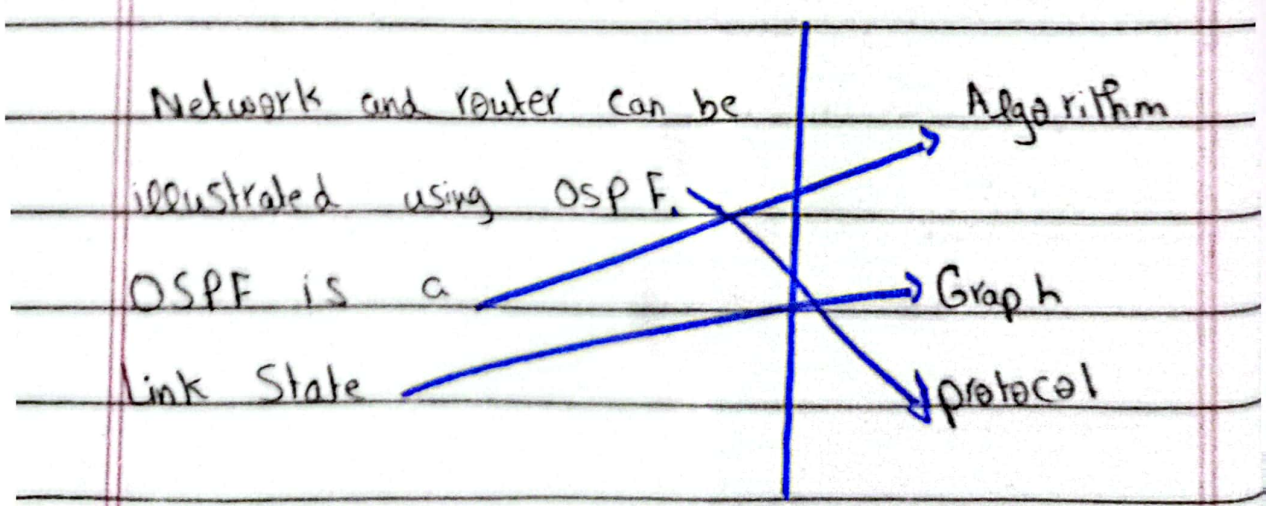
Answer:

"Fragmentation": one technique is to limit datagram size to smallest MTU of any network. IP uses datagram fragmentation can be split into ~~two~~ pieces to fit ~~into~~ network with small MTU. Router detect datagram large then network MTU and then it split into pieces and each pieces is smallest outbound network MTU.

Question:29

Match the following colum.?

Answer:



Date:

Question: 30

Network engineer wants to implement an ARP technique in which protocol address must be implemented from hardware address name of the technique.?

Answer:

Proxy ARP

Question: 31

Consider a large scale computer network where update or change made manually in routing table. you are required to mention the name of internet routing technique. which is being used in this organization.?

Answer:

Static Routing.